# 💳 PennyPath — Updated PRD (June 2025)

## 📌 tl;dr

PennyPath is a mobile-first personal finance iOS app (built in SwiftUI) focused on helping users manage credit-based financial products. The app supports tracking of credit cards, loans, BNPL agreements, budgeting, business expense tagging, and cash flow forecasting.

This updated PRD reflects progress to date and refines the core structure to enable accurate time-based analytics and future expansion.

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## 🌟 Goals

### Business Goals

- Deliver a functional MVP in SwiftUI for iOS with Firebase backend
- Build a credit-first finance tracking experience
- Implement smart forecasting, budgeting, and alerting
- Show debt reduction and upcoming obligations clearly

### User Goals

- Track real-time balances across credit products
- Add historical and future transactions with accurate balance tracking
- Schedule BNPL and recurring credit payments
- Set monthly budgets and receive cash flow alerts
- Tag and organize expenses by event or business purpose
- Define custom BNPL repayment plans to reflect provider-specific terms

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## ✅ Progress to Date (as of June 2025)

- ✅ **Firebase Auth Integration Completed**
  - User can sign up and log in using email/password
  - Basic Firestore user profile created on sign up
  - Navigates to placeholder home/dashboard on login
- ✅ **Account Model and Add Account View Implemented**
  - User can add a new account with:
    - `name` (e.g., Barclaycard)
    - `type` (credit card, loan, BNPL, other)

    - `opening_balance` (user-defined)
    - `opening_balance_date` (new refinement)
    - `alert_threshold` (optional low-balance warning)
  - `current_balance` is initialized to match `opening_balance`
  - Firestore structure and view models updated accordingly

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## 🔧 Updated Firestore `accounts/{accountId}` Schema

```
{
  "name": "Barclaycard",
  "type": "credit_card",
  "opening_balance": 825.00,
  "opening_balance_date": "2025-05-01",
  "current_balance": 1175.00,
  "alert_threshold": 50
}
```

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## 📅 Upcoming Feature: BNPL Plan Architecture

### New Collection: `bnpl_plans/{planId}`

Allow users to define reusable BNPL repayment plan templates.

```
{
  "provider": "Zilch",
  "plan_name": "Zilch Pay over 6 Weeks",
  "fee_type": "percentage",           // or "flat" or "none"
  "fee_value": 4.99,                   // 4.99% or £4.99
  "installments": 4,
  "payment_frequency": "biweekly",
  "initial_payment_percent": 25,
  "linked_account_id": null           // optional default
}
```

Plans can be selected when creating BNPL transactions, allowing consistent scheduling logic and customizable provider behavior.

### BNPL Transaction Example (Referencing Plan)

```
{
  "accountId": "bnpl_zilch_01",
  "amount": 120.00,
  "bnpl": true,
  "bnpl_plan_id": "plan_zilch6wk",
  "initial_payment": 30.00,
  "fee_applied": 4.99,
  "linked_account_id": "monzo_main",
  "scheduled_payment_ids": ["sp_01", "sp_02", "sp_03"],
  "date": "2025-06-15"
}
```

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## 📊 Prompt Guidelines for Next AI Steps

When continuing development with coding assistants, provide this updated PRD and specify the next task explicitly, e.g.,:

> "Use this PRD as the project context. Now build the View Accounts screen in SwiftUI that reads account data from Firestore and shows a summary list of accounts with name, type icon, and balance."
>
> "Use this PRD and help me build the BNPL Plan creation form, allowing a user to define and save a repayment plan with fee type, installments, and payment frequency."
>
> "When adding a BNPL transaction, allow the user to select an existing plan or create a new one inline. Apply the plan logic to schedule repayments and update the BNPL account balance."

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## 📈 Remaining MVP Features

- View accounts from Firestore
- Transactions (add/edit/delete)
- BNPL plan creation + transaction scheduling
- Budgets by category + budget progress
- Cash flow forecast + alerts
- Event tagging + business expense receipts
- Home dashboard
- Final polish & test